Review of Related Literature and Studies

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Course Title

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Due Date

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This chapter reviews the literature that grounds the proposed benchmarking of multitask optimization (MTO) on Taglish disaster tweets. The relevant work spans three areas: crisis informatics, MTO, and code-switched text processing. No single area establishes whether specialized balancing benefits triage on code-switched text, so the review covers the three areas in sequence. It arranges prior work by computational subconcept, algorithmic paradigm, and empirical bottleneck rather than by geographical origin.

Section 2.1 reviews crisis informatics corpora, joint-modeling evidence for entity, intent, and urgency tasks, cross-event and cross-lingual generalization, and the computational, statistical, and methodological standards the benchmark adopts. Section 2.2 then turns to MTO. It analyzes the gradient-surgery and dynamic loss-weighting families, the optimizer confound, and the controlled comparisons questioning whether specialized balancing outperforms tuned static linear scalarization. Section 2.3 narrows to the target domain, covering code-switching challenges and measurement, multitask learning on code-switched text, multilingual encoders, and Philippine language resources, including local disaster response studies. Section 2.4 consolidates the key studies in the Algorithmic Literature Taxonomy Matrix, which positions them against the proposed research. Section 2.5 closes with a three-step synthesis that derives the research gap and maps the study's contributions onto it.

Database searches conducted as of 2026 on MTO, crisis informatics, and code-switching identified the sources. These searches drew primarily on peer-reviewed conference and journal publications indexed in the ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, and the Association for Computational Linguistics (ACL) Anthology. Within this set, the crisis informatics and code-switching sources that established the evaluation domain date mostly from the last five years. The review retained older studies only if they defined a benchmarked algorithm, a candidate encoder, a methodological and statistical protocol, or a foundational corpus or metric.

2.1 Joint Multitask Modeling and Computational Efficiency in Crisis Informatics

Section 2.1 traces the crisis informatics arc from annotated corpora through joint triage models to cross-event generalization. It then sets the computational, statistical, and methodological standards the benchmark adopts.

2.1.1 Evolution of Crisis Informatics Corpora and Models

Both the corpus track and the model track begin with a demonstration that disaster tweets can be annotated and classified at scale. Imran et al. (2016) collected messages during 19 crises between 2013 and 2015 and released annotations for approximately 50,000 of them. The authors also published word embeddings trained on 52 million crisis-related tweets. The annotation schemes incorporated input from formal response agencies such as the United Nations Office for the Coordination of Humanitarian Affairs. The resulting nine categories included “caution and advice,” “displaced people and evacuations,” and “infrastructure and utilities damage.” Naive Bayes, random forest, and support vector machine classifiers reached an area under the curve of at least .80 for most categories, versus .50 for random classification. The exception was the smallest class, “missing, trapped, or found people.” The curation pipeline also flagged out-of-vocabulary terms such as slang, abbreviations, and misspellings, a noise thread that Section 2.3 develops.

Although disaster corpora continued to proliferate, disparate annotation schemes prevented direct cross-dataset comparisons. Alam, Sajjad, et al. (2021) documented that gap and responded with CrisisBench, consolidating eight human-annotated datasets into two corpora of 166,098 and 141,533 tweets. The curation pipeline eliminated three forms of duplication, pairing exact string matching with cosine similarity filtering to prune retweets and copied posts. That filtering kept duplicate messages out of both training and test splits.

Corpus construction then scaled up under explicit agreement control. Alam, Qazi, et al. (2021) released the Human-Annotated Disaster Incidents Data (HumAID), sampling tweets from a pool of about 24 million messages across 19 disaster events between 2016 and 2019. Crowd workers read 77,196 sampled tweets under three judgments per tweet, and a label was kept when two of the three agreed, a two-thirds (66%) agreement threshold. Combining the events for cross-event experiments triggered a further near-duplicate removal that left 76,466 tweets.

On these standardized benchmarks, fine-tuned encoders displaced the classical classifiers. CrisisBench's consolidated humanitarian task set the pattern: A fine-tuned RoBERTa model reached .872 on weighted F1-measure (F1), compared with .829 for the strongest convolutional-network baseline (Alam, Sajjad, et al., 2021). A fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model trailed RoBERTa by only .012 F1 on the same task. Evaluations on HumAID showed the same ordering across events. RoBERTa reached .781 weighted F1 on the combined split, whereas the support vector machine, random forest, and fastText classifiers stayed at or below .712 (Alam, Qazi, et al., 2021). Two cautions qualify these numbers. Weighted F1 averages per-class scores with class-frequency weights, so majority classes dominate the total; the benchmark adopted it precisely to factor in class imbalance. The two benchmarks differ in corpora, label sets, and splits, so their numbers support separate conclusions rather than one ranking.

Pre-training itself then moved into the crisis domain. Lamsal et al. (2024) trained CrisisTransformers on more than 15 billion tokens drawn from over 30 crisis events and evaluated the models on 18 crisis-specific public datasets. The domain-adapted models outperformed strong baselines across all of those datasets on classification, and the best sentence encoder improved the state of the art by 17.43%. The pre-training run, however, occupied six A100 GPUs for two months, a cost profile that only a cluster supports.

The current frontier couples triage with generative large language models, and it requires cluster computing. Yin et al. (2026) instruction fine-tuned a large language model for multilabel disaster classification and reported 63.8% overall accuracy compared with 4.27% for zero-shot prompted baselines, a 15-fold margin. Low-Rank Adaptation tuning used a 3-GPU A40 node, and full-parameter tuning ran on a 16-GPU A100 node. Zguir et al. (2025) extended triage toward actionable requests and offers through a hierarchical taxonomy and query-specific few-shot learning, evaluating on 300 manually annotated tweets alongside 1,346 synthetic examples. Results of this generation therefore rest on a 300-tweet manually labeled core and on hardware beyond a local workstation.

Across both tracks, three absences mark the space the proposed benchmark enters. First, every corpus in the line stopped at document-level labels; none paired token-level entity spans with concurrent sequence-level intent and urgency labels. Second, the systems behind the strongest results trained on clusters or called cloud services. Third, the corpus benchmarks reported accuracy and F1 but no inference latency or memory. These three absences, not any single result, define the gap, and Section 2.5 returns to them.

2.1.2 Joint Modeling of Token-Level and Sequence-Level Triage Tasks

The joint-modeling question predates crisis informatics by two decades. Caruana (1997) defined multitask learning as inductive transfer that improves generalization by using the domain information in the training signals of related tasks. Models learn tasks in parallel over a shared representation, so what the model learns for one task helps the others. Baxter (2000) supplied the theory: A hypothesis space that performs well on sufficiently many related tasks will also perform well on novel tasks from the same environment.

The nearest structural precedent comes from spoken-language understanding. Weld et al. (2022) surveyed the joint models behind conversational agents and chatbots. The standard approach trains models for intent detection over the whole utterance together with slot filling over its individual tokens. This pairing exhibits the structural configuration that the proposed triage formulation requires: one token-level task alongside sequence-level tasks.

Prior crisis informatics studies applied joint modeling only partially. Wang et al. (2021) fine-tuned one BERT model on the TREC Incident Streams (TREC-IS) crisis corpus to predict information types and priority levels together. The loss weight remained fixed at .50, weighting both task objectives equally. Seeberger and Riedhammer (2022) combined entity-masked language modeling with hierarchical multilabel classification on the same TREC-IS corpus under its 2020B split. The combined design gained up to 10 percentage points of F1 on actionable information types, and entity masking reduced overfitting to in-domain events.

Prior crisis studies have not evaluated this concurrent pairing. No reviewed study evaluated token-level entity extraction together with intent and urgency classification on the same disaster tweets, and none examined the optimization strategies such architectures demand. Whether specialized balancing resolves that shared loss is the question Section 2.2 takes up.

2.1.3 Cross-Event and Cross-Lingual Generalization

Evaluation practice moved toward held-out events in the same years. Ray Chowdhury et al. (2020) trained multilingual BERT with Manifold Mixup and tested on four datasets, three of which came from disasters absent from the training data. On the English-only evaluation, the model reached an F1 of 79.36% on the Philippines Flood event. Zero-shot transfers scored 81.33% on French, 75.44% on Italian, and 85.26% on Spanish. Per-class evaluation on the multilingual mixed set showed the lowest score on the caution and advice class, at 70.3% F1.

Cross-lingual evaluations also quantified the performance drop incurred during low-resource transfer. Sarioglu Kayi et al. (2020) trained urgency classifiers on English crisis tweets and released evaluation sets for Sinhala and Odia. The ensemble classifier scored 76.5% F1 on English but dropped to 63.5% on Sinhala and 62.6% on Odia when transferred without target-language training. The authors traced part of the noise to annotation because general status updates and critical help requests both carried the urgent label.

Recent work treated generalization failure itself as a modeling problem. Seeberger et al. (2025) showed that crisis classifiers relied on spurious event-specific cues such as locations and hashtags. The authors proposed a causal debiasing framework with domain-specific experts. Across three disaster classification tasks under temporal splits to unseen events, the approach outperformed the next-best baseline by 1.1–1.9 percentage points and significantly improved the BERT baseline. This line of evidence justifies the leave-one-event-out (LOEO) evaluation of the proposed benchmark, which holds out one typhoon at a time.

2.1.4 Complexity Analysis, Statistical Rigor, and Design Science Methodology

The benchmark's complexity axis derives from a structural property of shared encoders. One forward pass over the encoder serves all three triage tasks, whereas three single-task systems incur the encoder computation three times per tweet. Inference latency and memory therefore become measurable benchmark axes in their own right. The MTO landscape is also consolidated enough to compare: LibMTL packaged 13 optimization strategies and eight architectures behind one reproducible PyTorch interface (Lin & Zhang, 2023).

Single reported scores carry a variance risk that single runs hide. Reimers and Gurevych (2017) retrained sequence-tagging networks with identical hyperparameters but different random seeds. Maximum score differences reached 2.59 F1 points on the Conference on Computational Natural Language Learning (CoNLL) 2003 named entity recognition benchmark. The gap reached 8.23 F1 points on the Automatic Content

Extraction (ACE) 2005 Entities task. Selecting a run by its development score was also unreliable, with Spearman's rank correlation between development and test scores at only $\rho = .23$ in the reported case. The proposed benchmark therefore trains every configuration under multiple seeds and treats single scores as insufficient evidence.

Comparison across many systems and datasets adds its own statistical constraints. Demšar (2006) showed that parametric tests rest on assumptions that comparisons across multiple datasets violate. The recommended alternative is the Friedman test with corresponding post hoc procedures, using the Nemenyi procedure for all-pairwise comparisons, together with the Iman–Davenport correction for the conservative Friedman statistic. The proposed benchmark adopts this nonparametric protocol for its method comparisons.

The last standard concerns the study's form. Peffers et al. (2007) formalized design science research methodology as a process model of six activities in a nominal sequence, a framework for research that creates and evaluates artifacts. The proposed study follows that framing: The benchmark corpus and the prototype triage service are its design artifacts. No reviewed crisis study combines latency and memory measurement, multi-seed evaluation, nonparametric comparison, and artifact framing in one benchmark, which is the standards gap that Section 2.5 assembles.

2.2 Multitask Optimization Dynamics

Training one network on several tasks turns optimization into a balancing problem. Shared parameters serve every task loss at once, so an update that helps one task can hurt another. Two algorithmic answers dominate the response literature. Gradient surgery edits the shared update direction, whereas dynamic loss weighting adapts how much each loss counts. A third response acts on the architecture, using attention modules that modulate the shared representation across tasks (S. Liu et al., 2019), and the proposed benchmark holds that side fixed. Sener and Koltun (2018) supplied the frame, casting multitask learning as Pareto-optimal search, and their Multiple Gradient Descent Algorithm–Upper Bound (MGDA-UB) approximation anchored the optimization line this section reviews.

The surgery family begins with projecting conflicting gradients (PCGrad), whose authors named the obstacle precisely. Yu et al. (2020) defined two gradients as conflicting when they point away from each other, that is, when their cosine similarity is negative. Their repair projected a task's gradient onto the normal plane of any conflicting gradient before the update, a general procedure they called "gradient surgery," whose conflicting-gradient instance is PCGrad. On a series of challenging multitask supervised and multitask reinforcement learning problems, the approach yielded substantial gains in efficiency and performance.

Successors then reshaped the surgery idea along three main lines, with two additional context methods. B. Liu et al. (2021) introduced Conflict-Averse Gradient Descent (CAGrad), which maximized worst-case improvement near the average gradient while minimizing average loss, with a convergence guarantee earlier projection methods lacked. L. Liu et al. (2021) contributed Impartial Multi-Task Learning (IMTL-G), which rescaled per-task gradients in closed form so that the aggregated gradient carried equal projections onto every task. Nash multitask learning (Nash-MTL) replaced pairwise repair with a principled global rule, adopting the Nash bargaining solution from cooperative game theory (Navon et al., 2022). Two context methods completed the family.

Independent Component Alignment for Multi-Task Learning (Aligned-MTL) aligned principal components via singular value decomposition (Senushkin et al., 2023). Similarity-aware momentum gradient surgery (SAM-GS) equalized gradients when magnitude similarity signaled conflict and modulated momentum otherwise (Borsani et al., 2025).

The weighting family tunes scalar loss weights rather than editing gradient directions directly. Kendall et al. (2018) balanced loss functions by weighting each task with its homoscedastic (task-dependent) uncertainty, which reduces the influence of noisy tasks during optimization. Gradient Normalization (GradNorm) learned the weights by steering per-task gradient norms toward targets set by relative inverse training rates (Chen et al., 2018). Fast Adaptive Multitask Optimization (FAMO) reduced the memory and time cost of gradient-manipulation methods by equalizing relative loss-improvement rates from loss history without storing per-task gradients (B. Liu et al., 2023).

Controlled comparisons then tested whether specialized balancing improves results. Kurin et al. (2022) defended unitary scalarization, training that sums the task losses. Coupled with single-task regularization and stabilization, it matched complex multitask optimizers or improved upon them in popular supervised and reinforcement learning settings. Most of those optimizers require per-task gradients and introduce memory, runtime, and implementation overhead that scales with task count. Xin et al. (2022) reached the same verdict at scale, finding no improvements across language and vision tasks beyond what traditional optimization approaches achieve. Elich et al. (2024) questioned the mechanism itself, finding no evidence that angular gradient conflict is unique to multitask learning and locating the distinguishing factor in gradient magnitude.

Gama and Grassi (2025) added the boundary condition: Specialized optimizers are unnecessary on simple problems, and their value re-emerges as task counts and feature-space demands grow.

One optimizer confound affects the multitask comparison dispute. Researchers have often paired balancing rules with different optimizers, so a reported gain may reflect the optimizer rather than the balancing rule. Loshchilov and Hutter (2019) introduced AdamW, decoupling weight decay from the gradient-based update. Elich et al. (2024) derived a partial loss-scale invariance under mild assumptions, showing that loss scaling cancels out in head parameters while still affecting the shared backbone. The procedural remedy is direct: Match budgets, fix the optimizer, and tune baselines as carefully as evaluated methods (Elich et al., 2024; Kurin et al., 2022; Xin et al., 2022).

The claims of the surgery and weighting families and the counterclaims from the controlled comparisons therefore remain unresolved in the general domain apart from the complexity boundary condition. Researchers on neither side have tested that dispute on code-switched, low-resource text, and no reviewed study in that literature has compared optimization strategies under controlled conditions. That untested intersection is the evidence gap, and Section 2.3 examines the code-switched literature behind that gap.

2.3 Code-Switched Text Processing and Low-Resource Philippine NLP

Standard natural language processing (NLP) pipelines degrade on code-switched text because their underlying architectures assume a single language per document (Winata et al., 2023). Subword tokenization introduces the earliest computational bottleneck, fragmenting morphologically hybrid words into excessive subword pieces.

Across the parallel FLORES-200 benchmark, translated content required up to 15 times more tokens than English, a disparity that persisted across multilingual tokenizers (Petrov et al., 2023). For the proposed Taglish corpus, tokenization therefore requires deliberate control as an experimental design variable.

Beyond tokenization disparities, code-switching also varies in intensity across individual utterances. Gambäck and Das (2016) refined the Code-Mixing Index to quantify this utterance-level switching, providing the metric this study averages alongside its rule-based language profiles. Downstream Taglish evaluation has only recently begun: Adoptante et al. (2025) evaluated open-source large language models on Tagalog–English retrieval-augmented generation, finding that Mistral and SeaLLM generated the most grounded answers. Although generative models anchor those initial benchmarks, latency-sensitive crisis triage instead favors compact discriminative encoders.

The encoder lineage behind the candidate backbones runs in three waves. Devlin et al. (2019) introduced BERT, pretraining deep bidirectional transformers for language understanding. Sanh et al. (2019) distilled BERT into a model 40% smaller that retained 97% of BERT's language understanding capabilities while running 60% faster. Conneau et al. (2020) scaled masked pretraining to 100 languages with XLM-RoBERTa (XLM-R), lifting cross-lingual natural language inference accuracy by 14.6% over multilingual BERT on average. Multilingual BERT and XLM-R serve as two of the study's nine candidate backbones. XLM-R demonstrated particular strength on low-resource languages, providing the empirical rationale for its selection.

The second wave specialized the pretraining recipe itself. Chung et al. (2021) rethought embedding coupling, showing that decoupled embeddings provided increased modeling flexibility and improved parameter-allocation efficiency. Clark et al. (2022) removed the tokenizer altogether, pretraining Canine directly on characters as a tokenization-free encoder. Zhang et al. (2023) turned to the domain itself, pretraining TwHIN-BERT on 7 billion tweets in 100 languages to model short, noisy, user-generated text.

The most recent wave scales vocabulary, architecture, and regional coverage. Liang et al. (2023) overcame the vocabulary bottleneck with XLM-V, a 1-million-token vocabulary model that outperformed XLM-R on every task tested. Marone et al. (2026) trained mmBERT on 3 trillion tokens across more than 1,800 languages, adding an inverse mask-ratio schedule among its novelties. AI Singapore (2026) brought the lineage to Southeast Asia with SEA-LION ModernBERT, which covers 13 languages and uses a Gemma 3 tokenizer. Its Filipino coverage and encoder-only design make it one of the study's nine candidate backbones.

The review's one direct precedent for multitask learning on code-switched text came from Algerian text and suggested caution. Adouane and Bernardy (2020) jointly learned four tasks progressively on noisy, user-generated text in Algerian, a low-resource non-standardized Arabic variety. The tasks were code-switch detection, named entity recognition, spell normalization and correction, and sentiment identification. Multitask learning proved beneficial for some tasks in particular settings. The effect of each task on another, the order of the tasks, and the size of the task with more data all mattered. For a study whose treatment variable is the optimization strategy, that conditionality is the warning: Gains observed elsewhere cannot be assumed to hold on code-switched text.

Parallel to these architectural developments, Philippine-language resources have matured. Cruz and Cheng (2022) released TLUnified, a large-scale Filipino pretraining corpus, alongside RoBERTa Base and Large models that outperformed prior baselines. The released RoBERTa Base checkpoint serves as the study's RoBERTa-Tagalog Base candidate. Africa et al. (2025) introduced meta-pretraining with first-order model-agnostic meta-learning for zero-shot cross-lingual named entity recognition in Tagalog and Cebuano. Montalan et al. (2024) contributed Kalahi, a 150-prompt handcrafted suite that evaluated large language models on Filipino culture.

Tagalog-specific tooling and Taglish datasets followed. Miranda (2023b) released a Tagalog named entity recognition dataset annotated for person, organization, and location, with agreement of .81 by Cohen's kappa on all tokens. The calamanCy toolkit provided Tagalog pipelines for part-of-speech tagging, parsing, and named entity recognition (Miranda, 2023a). Herrera et al. (2022) collected 21,150 Tagalog–English tweets for TweetTaglish, and the best model identified language distributions at r^2 values of .797–.909.

The local disaster-response strand exists but remains single task and single event. Ermino et al. (2022) trained a lightweight feedforward network on 2,010 urgent and 2,010 nonurgent tweets from Typhoon Ulysses (Vamco). The model reached 99.17% validation accuracy on the 120-tweet set, with 97.67% sensitivity and 100% specificity on the separate 100-tweet confusion-matrix analysis. No study in this strand offers a Taglish triage corpus with concurrent token-level and sequence-level labels. That absence is the contextual gap, and Section 2.5 assembles it with the others.

2.4 Algorithmic Literature Taxonomy Matrix

This section consolidates the review into the Algorithmic Literature Taxonomy Matrix. Table 1 summarizes the key related studies under the five institutional columns: author and year, algorithm or model, dataset or testbed, evaluated metrics, and identified limitation. The row order follows the preceding sections, from crisis benchmarks through multitask optimization methods to candidate encoders and low-resource precedents. Every row contrasts a design-shaping model or empirical baseline with the proposed approach: a benchmarked optimization method, a scalarization baseline, a candidate encoder, or an empirical task precedent. The final row states the proposed study against these entries.

Table 1

Algorithmic Literature Taxonomy Matrix

Author and year	Algorithm or model	Dataset or testbed	Evaluated metrics	Identified limitation
Alam, Sajjad, et al. (2021)	Consolidated benchmarks with CNN, fastText, and transformer baselines	8 human-annotated datasets merged into 166,098 and 141,533 tweets	Weighted F1, RoBERTa .872 humanitarian and .883 informativeness	Document-level labels; no latency or memory reported
Alam, Qazi, et al. (2021)	Random forest, SVM, fastText, BERT, DistilBERT, RoBERTa, and XLM-R baselines	HumAID, 77,196 tweets across 19 events, 76,466 after cross-event deduplication	Weighted F1 on the combined split, RoBERTa .781, non-transformer baselines at or below .712	Document-level labels; single task
Wang et al. (2021)	Transformer multi-task learning with encoder ensembles	TREC-IS 2018 through 2020A splits	Macro F1 on all and actionable types; accuracy	Fixed equal loss weighting at .50; optimization unexamined

Author and year	Algorithm or model	Dataset or testbed	Evaluated metrics	Identified limitation
Yu et al. (2020)	PCGrad, projecting conflicting gradients	MultiMNIST, Cityscapes, CelebA, CIFAR-100, NYUv2, and Meta-World RL	Efficiency and performance gains over single-task training	Per-task gradients; overhead scales with task count
B. Liu et al. (2021)	CAGrad, conflict-averse gradient descent with convergence guarantee	Multi-Fashion+MNIST, NYUv2, Cityscapes, MT10 and MT50 RL, CIFAR-10 semi-supervised	Improved over prior gradient-manipulation methods	Per-task gradients; overhead scales with task count
L. Liu et al. (2021)	IMTL, the hybrid of IMTL-G gradient balance and IMTL-L loss balance	Cityscapes, NYUv2, and CelebA	Outperformed existing loss-weighting methods	Per-task gradients; overhead scales with task count
Navon et al. (2022)	Nash-MTL, Nash bargaining solution	QM-9, NYUv2, Cityscapes, and MT10	State-of-the-art MTL performance with convergence guarantees	Per-task gradients; overhead scales with task count
Kendall et al. (2018)	Homoscedastic uncertainty weighting of task losses	Cityscapes depth, semantic, and instance segmentation	Outperformed separately trained single-task models	Vision tasks only; no text evaluation
Chen et al. (2018)	GradNorm, gradient-norm balancing toward training-rate targets	Synthetic 2- and 10-task regression; NYUv2 variants	Improved accuracy and reduced overfitting	Requires per-task gradient norms; vision-era benchmarks

Author and year	Algorithm or model	Dataset or testbed	Evaluated metrics	Identified limitation
Liu et al. (2023)	FAMO, loss-improvement-rate balancing from loss history	NYUv2, Cityscapes, QM-9, CelebA, and MT10	Matched gradient-manipulation methods at lower time and memory	Dependent on a regularization parameter; no text tasks
Kurin et al. (2022)	Unitary scalarization with single-task regularization and stabilization	MultiMNIST, CelebA, Cityscapes, and Meta-World	Matched or improved over multitask optimizers	Coverage cannot exclude settings where scalarization loses
Xin et al. (2022)	Tuned scalarization sweep across static task weights	Multilingual translation, Cityscapes, CelebA, and MultiMNIST	No MTO gains beyond tuned scalarization	Frontier search computationally prohibitive
Devlin et al. (2019)	BERT, bidirectional masked pretraining	BooksCorpus and Wikipedia; GLUE, SQuAD, and SWAG	State of the art on 11 NLP tasks; MultiNLI accuracy 86.7% (GLUE score 80.5%)	English benchmarks; no multilingual or crisis evaluation
Conneau et al. (2020)	XLM-R, multilingual masked pretraining at scale	CC-100, 2.5 TB filtered CommonCrawl, 100 languages	+14.6% XNLI accuracy over multilingual BERT	Curse of multilinguality capacity trade-off
Adouane & Bernardy (2020)	Progressive joint learning of 4 tasks with shared CNN encoders	4 Algerian code-switched datasets for detection, normalization, sentiment, and NER	Per-task performance in joint and single-task settings	Benefits conditional on pairing, order, and data size
Cruz & Cheng (2022)	Tagalog RoBERTa Base and Large on TLUnified	TLUnified; hatespeech, dengue, and news benchmarks	Outperformed BERT and Electra by roughly 4–5 points	Large variant may need more pretraining data

Author and year	Algorithm or model	Dataset or testbed	Evaluated metrics	Identified limitation
Ermino et al. (2022)	Feedforward artificial neural network	4,020 Typhoon Ulysses tweets, 2,010 urgent and 2,010 nonurgent	99.17% validation accuracy; 97.67% sensitivity; 100% specificity	Single task and single typhoon
The proposed study (2026)	Hard-sharing multilingual encoder with 3 task heads under 10 optimization configurations against single-task, equal-weight, and tuned static linear scalarization baselines	Multi-Task Corpus of Taglish Disaster Tweets, leave-one-event-out folds across 4 typhoons	Task-level macro and weighted F1, latency, memory, and Friedman, Iman–Davenport, and Nemenyi tests	4 typhoons, Taglish only, encoder-only scope

Note. Limitations are stated by the original authors or identified by this review in Sections 2.1 through 2.3. MTL = multitask learning; MTO = multitask optimization; NER = named entity recognition; RL = reinforcement learning; F1 = F1-measure.

The matrix consolidates the pattern that the synthesis assembles. Across the matrix, the identified limitations converge: the reviewed studies leave the triage tasks unpaired, the optimization comparison untested on code-switched text, and the deployment cost unmeasured. The proposed study row takes each absence as a design target. Section 2.5 assembles the gaps and the study’s contributions.

2.5 Synthesis of Related Literature and Studies

Two empirical convergences emerge from the reviewed literature. First, fine-tuned encoders displaced classical classifiers on standardized crisis benchmarks, demonstrating consistent transformer advantages over non-transformer baselines. On CrisisBench's consolidated humanitarian task, fine-tuned RoBERTa achieved .872 weighted F1, compared with .829 for the strongest convolutional baseline (Alam, Sajjad, et al., 2021). On HumAID's combined split, RoBERTa reached .781 weighted F1 whereas non-transformer classifiers stayed at or below .712 (Alam, Qazi, et al., 2021). Second, multi-task learning research produced standardized architectures and benchmarking suites. Prior spoken-language systems established the structural precedent of pairing token-level sequence labeling with sentence-level classification (Weld et al., 2022). Lin and Zhang (2023) consolidated 13 optimization strategies and eight architectures into a single reproducible multi-task learning library. Evaluation conventions also carry forward: Weighted F1 accounts for class imbalance, macro F1 exposes minority-class performance, and the proposed study reports both metrics per task.

These convergences establish the baseline architecture but leave four critical research gaps unaddressed. The first gap is contextual. No reviewed corpus provides concurrent token-level entity spans alongside intent and urgency labels for Taglish disaster tweets. Furthermore, the existing Philippine disaster literature remains restricted to single-task, single-event classification (Ermino et al., 2022). The second gap is evidentiary. Controlled comparisons show that the dispute over specialized multitask optimization remains unresolved in the general literature (Kurin et al., 2022; Xin et al., 2022). Crucially, no empirical study has evaluated specialized multitask balancing on low-resource, code-switched text. The third gap is practical. No reviewed crisis study reported inference latency or memory alongside accuracy and F1 metrics. The fourth gap is methodological. Prior crisis studies omitted multi-seed evaluation, nonparametric significance testing, deployment-cost benchmarking, and formal artifact framing. These absences compound one another: Unannotated data preclude multitask optimization research, while unmeasured computational costs prevent deployment in disaster response.

The proposed study addresses each absence with a specific contribution. Developing the Multi-Task Corpus of Taglish Disaster Tweets with concurrent token- and sequence-level labels closes the contextual gap. Benchmarking 10 optimization configurations against single-task, equal-weight, and tuned static linear scalarization baselines closes the evidence gap. Measuring inference latency and memory alongside macro and weighted F1 closes the practical gap. Evaluating multi-seed leave-one-event-out folds with Friedman, Iman–Davenport, and Nemenyi tests closes the methodological gap. The investigation follows design science research methodology (Peffers et al., 2007). The curated corpus and the prototype triage service serve as the primary design artifacts, evaluated against the operational requirements identified across the literature. Chapter 3 presents the methodology used to construct and evaluate both artifacts.

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